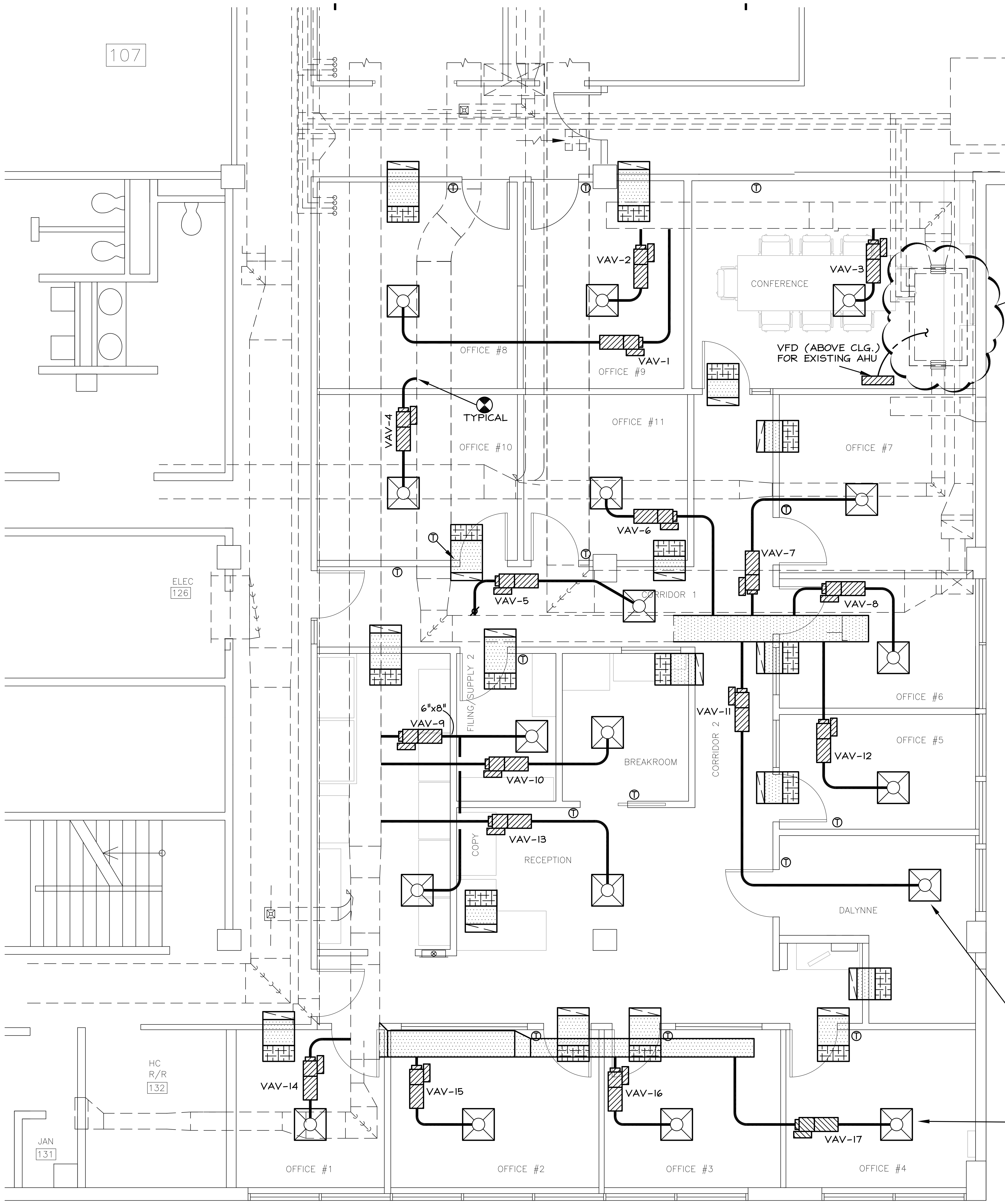


CONTRACTOR INDICATED
A PROPOSAL REQUEST TO REPLACE EXISTING AIR HANDLING UNIT WITH NEW (MATCHING CAPACITY OF EXISTING) 2 HP VARIABLE SPEED CAPABLE MOTOR- PENDING.

NOTE: INTERLOCK VAV CONTROLS TO EXISTING AHU & VFD. EXPAND EXISTING JCI DDC SYSTEM FOR VAV OPERATION AS REQUIRED. COORDINATE ALL REQUIREMENTS PRIOR TO FURNISHING A PROPOSAL.

VARIABLE AIR VOLUME BOX SCHEDULE

COOLING					HEATING								
UNIT No	CFM COOLING		INLET SIZE	MAX SP ACROSS BOX	HEATING CAPACITY BTUH	CFM	AIR TEMP (° F)		KW STRIP		ELECTRIC SERVICE	REMARKS
	MIN	MAX					EAT	LAT	KW	STAGES		
VAV-1	45	125	4"ø	.25"	4,320	100	55	95	2.0	1	480/1/60	TITUS DESV-4 OR APPROVED EQUIVALENT
VAV-2	45	110	4"ø	.25"	3,890	90	55	95	2.0	1	480/1/60	TITUS DESV-4 OR APPROVED EQUIVALENT
VAV-3	85	340	6"ø	.25"	11,665	270	55	95	3.5	1	480/1/60	TITUS DESV-6 OR APPROVED EQUIVALENT
VAV-4	45	110	4"ø	.25"	3,890	90	55	95	2.0	1	480/1/60	TITUS DESV-4 OR APPROVED EQUIVALENT
VAV-5	45	105	4"ø	.25"	3,675	85	55	95	2.0	1	480/1/60	TITUS DESV-4 OR APPROVED EQUIVALENT
VAV-6	45	100	4"ø	.25"	3,455	80	55	95	2.0	1	480/1/60	TITUS DESV-4 OR APPROVED EQUIVALENT
VAV-7	45	150	4"ø	.25"	5,185	120	55	95	2.0	1	480/1/60	TITUS DESV-4 OR APPROVED EQUIVALENT
VAV-8	70	275	6"ø	.25"	9,505	220	55	95	2.8	1	480/1/60	TITUS DESV-6 OR APPROVED EQUIVALENT
VAV-9	45	100	4"ø	.25"	3,455	80	55	95	2.0	1	480/1/60	TITUS DESV-4 OR APPROVED EQUIVALENT
VAV-10	45	185	4"ø	.25"	6,480	150	55	95	2.0	1	480/1/60	TITUS DESV-4 OR APPROVED EQUIVALENT
VAV-11	55	225	6"ø	.25"	7,775	180	55	95	2.3	1	480/1/60	TITUS DESV-6 OR APPROVED EQUIVALENT
VAV-12	60	245	6"ø	.25"	8,425	195	55	95	2.5	1	480/1/60	TITUS DESV-6 OR APPROVED EQUIVALENT
VAV-13	55	215	4"ø	.25"	7,345	170	55	95	2.2	1	480/1/60	TITUS DESV-4 OR APPROVED EQUIVALENT
VAV-14	70	285	6"ø	.25"	9,940	230	55	95	3.0	1	480/1/60	TITUS DESV-6 OR APPROVED EQUIVALENT
VAV-15	110	440	6"ø	.25"	15,120	350	55	95	4.5	1	480/1/60	TITUS DESV-6 OR APPROVED EQUIVALENT
VAV-16	80	325	6"ø	.25"	11,230	260	55	95	3.3	1	480/1/60	TITUS DESV-6 OR APPROVED EQUIVALENT
VAV-17	90	365	6"ø	.25"	12,530	290	55	95	3.7	1	480/1/60	TITUS DESV-6 OR APPROVED EQUIVALENT

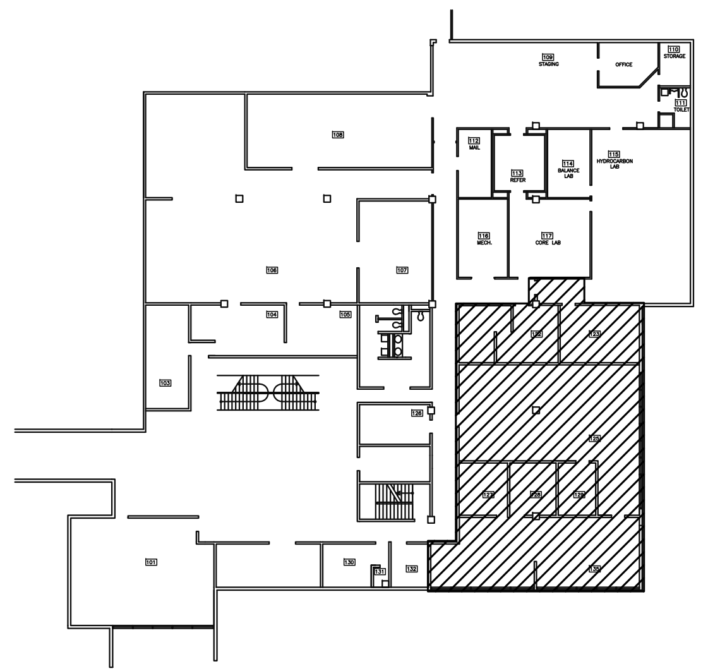
NOTE: CONTRACTOR SHALL PROVIDE SCALED DRAWINGS OF DOUBLE LINE VAV DUCT LAYOUTS AND ASSOCIATED DEMOLITION LAYOUT TO FACILITATE INSTALLATION OF VAV LAYOUT FOR THIS AREA AS PART OF BID ADDITIVE #1 WORK. COORDINATE REQUIREMENTS WITH ALL TRADES.

NOTE: REFER TO SHEET M1.0 FOR GRILLE TYPE AND REQUIRED MAXIMUM AIR FLOW (TYPICAL). RUN DUCT FULL SIZE BACK TO VAV BOX. TRANSITION TO BOX DISCHARGE OPENING AS REQUIRED.

NOTE:
REFER TO ARCHITECTURAL PLANS FOR ALL DIMENSIONS.

1ST FLOOR MECHANICAL PLAN- VAV SINGLE LINE LAYOUT -BID ADDITIVE #1

NOT TO SCALE



KEY PLAN
NOT TO SCALE

ASSOCIATED DESIGN GROUP, INC.
A PROFESSIONAL ENGINEERING COMPANY

CORPORATE OFFICE
3909 West Congress Street
Suite 201
Lafayette, Louisiana 70506
VOICE (337) 234-5710
FAX (337) 237-1467
EMAIL: adginc@adginc.org

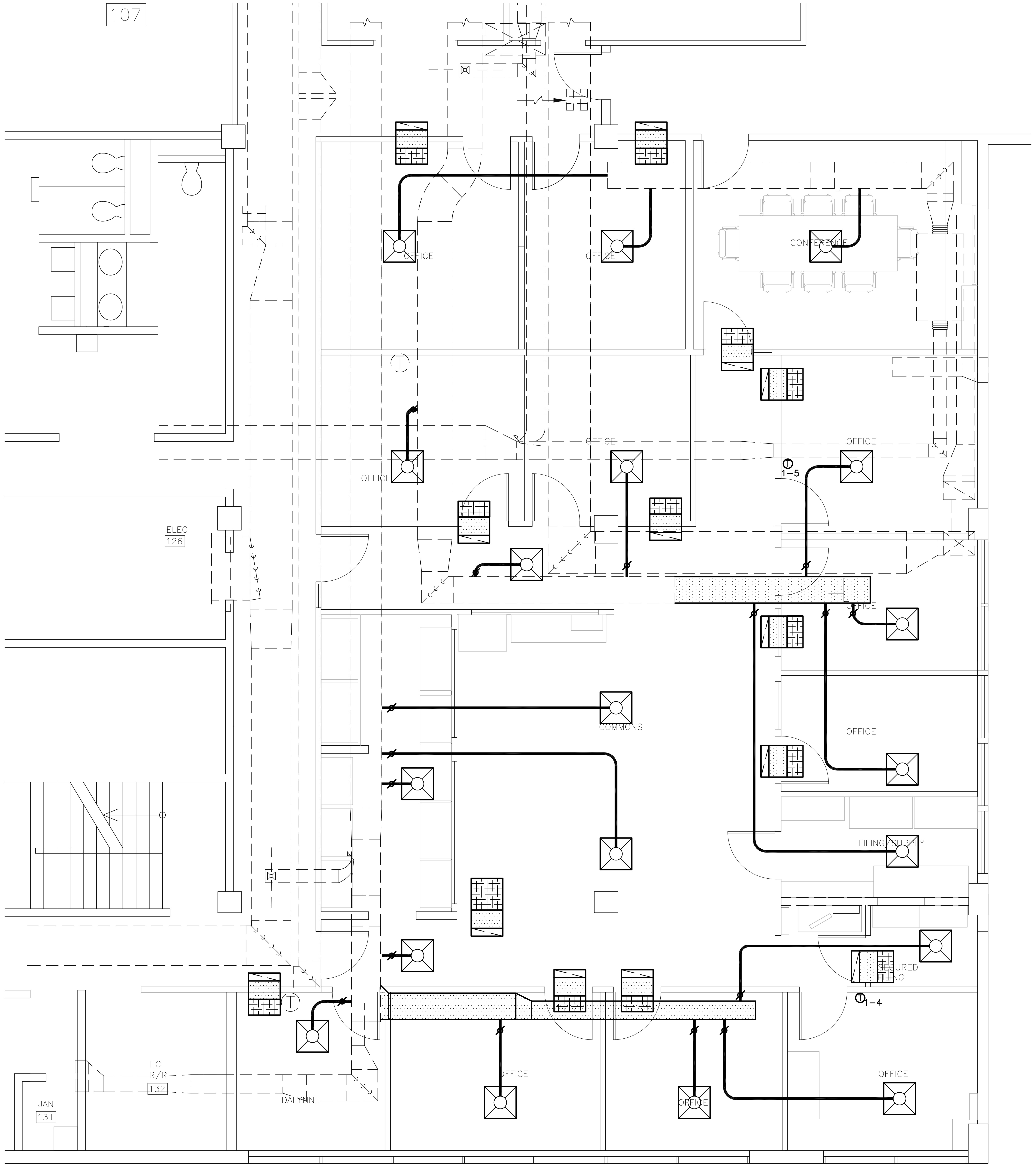
ADG Project No. 13124-P

UNITED STATES DEPARTMENT OF COMMERCE
NATIONAL OCEANIC & ATMOSPHERIC ADMINISTRATION
ATLANTIC OCEANOGRAPHIC AND METEOROLOGICAL LABORATORY
1ST FLOOR ADMINISTRATION RENOVATION

MIAMI, FLORIDA

HVAC REVISIONS

DATE ISSUED 04/12/16
REVISED Nov. 22, 2016
DRAWN BY JTP
CHECKED BY LWB
PROJECT NO. 13124-P
SHEET NO. M1.1
OF



1ST FLOOR ENLARGED MECHANICAL PLAN

SCALE 1/4"=1'-0"

HVAC GENERAL NOTES:

1. ALL WORK SHALL COMPLY WITH LOCAL, STATE, AND FEDERAL CODES TO THE SATISFACTION OF CODE AUTHORITIES HAVING JURISDICTION.
2. ALL WORK SHALL COMPLY WITH THE REQUIREMENTS OF NFPA 90A & 101, THE CLEAN AIR ACT AND THE AMERICANS WITH DISABILITIES ACT.
3. COORDINATE ALL CEILING DEVICES, DUCTWORK, ETC. WITH LIGHTING, STRUCTURE, ETC. THROUGH GENERAL CONTRACTOR. REFER TO ALL DRAWINGS, (STRUCTURAL, PLUMBING, ELECTRICAL, ARCHITECTURAL, ETC.). NOTIFY ARCHITECT/ENGINEER CONCERNING ANY CONFLICTS NOTED PRIOR TO BIDS FOR CLARIFICATION TO THE SATISFACTION OF THE BIDDER. REFER TO SPECIFICATIONS SECTION 15000 FOR REQUIREMENTS. REFER TO LATEST ARCHITECTURAL REFLECTED CEILING PLAN. COORDINATE ALL CEILING DEVICE LOCATIONS WITH CEILING GRID.
4. ALL SUPPLY AND RETURN DUCTWORK SHALL BE EXTERNALLY INSULATED. DUCT SIZES SHOWN ARE CLEAR METAL TO METAL. WRAP DUCTWORK EXTERNALLY.
5. HARDCAST ALL DUCTWORK JOINTS FOR AIRTIGHT SYSTEM. REFER TO DETAILS FOR DUCTWORK TAP, SPLITTER, BRANCH CONFIGURATION, ETC. INSTALL IN ACCORDANCE WITH SMACNA RECOMMENDATIONS AND INSTALLATION GUIDELINES.
6. PROGRAMMABLE THERMOSTAT SHALL REMAIN IN FAN "AUTO" MODE.

HVAC KEYNOTES:

◇1 RUN MAIN TRUNKLINES APPROXIMATELY AS SHOWN. COORDINATE EXACT LOCATION WITH STRUCTURE, PIPING, CONDUIT, DEVICES ABOVE CEILING, ETC. MAKE ALL REQUIRED OFFSETS, TRANSITIONS, ETC. (MAINTAIN EQUIVALENT DUCT FREE AREA IN TRANSITIONS). RUN DUCTWORK TO MINIMIZE OFFSETS, CHANGES IN ELEVATION, TRANSITIONS, ETC. COORDINATE WITH OTHER TRADES. (TYPICAL)

◇2 INSTALL MANUAL DAMPER AT DUCT TAP SPIN FITTING ABOVE LAY-IN CEILING (TYPICAL) AND BALANCE TO CFM INDICATED AT GRILLE. DUCT TAPS SHALL BE A MINIMUM OF 12" FROM END OF SUPPLY AIR DUCT. (TYPICAL)

◇3 INSTALL NEW 7 DAY PROGRAMMABLE THERMOSTAT AND CONTROL WIRING WHERE INDICATED.

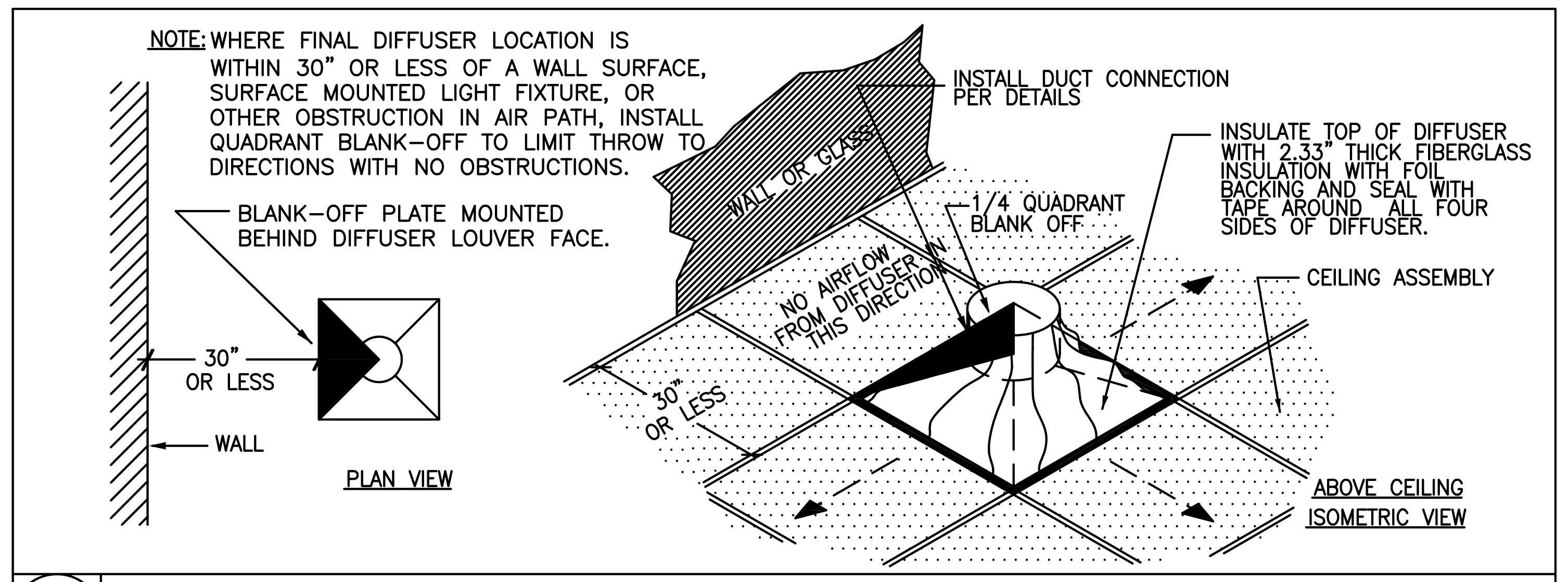
◇4 EXTEND EXISTING DUCTWORK AT SIZE INDICATED AS SHOWN.

◇5 CONNECT NEW DUCT TO EXISTING DUCT AS SHOWN. VERIFY EXACT LOCATION IN FIELD.

GRILLE SCHEDULE

SYMBOL	SIZE	SERVICE	LOCATION	FINISH	REMARKS
A	6"ø NECK	SUPPLY	CEILING	W.B.E.	TITUS TMSA OR APPROVED EQUAL (24"x24" LAY-IN T-BAR)
B	8"ø NECK	SUPPLY	CEILING	W.B.E.	TITUS TMSA OR APPROVED EQUAL (24"x24" LAY-IN T-BAR)
C	10"ø NECK	SUPPLY	CEILING	W.B.E.	TITUS TMSA OR APPROVED EQUAL (24"x24" LAY-IN T-BAR)
D	12"ø NECK	SUPPLY	CEILING	W.B.E.	TITUS TMSA OR APPROVED EQUAL (24"x24" LAY-IN T-BAR)
R *	24"x12"	RETURN	CEILING	W.B.E.	TITUS 50F OR APPROVED EQUAL (EGGCRATE GRILLE)

* SHEET METAL CONTRACTOR TO PROVIDE SHEET METAL CEILING BOX FOR ATTACHMENT OF DUCTWORK TO RA/EA AIR REGISTER.

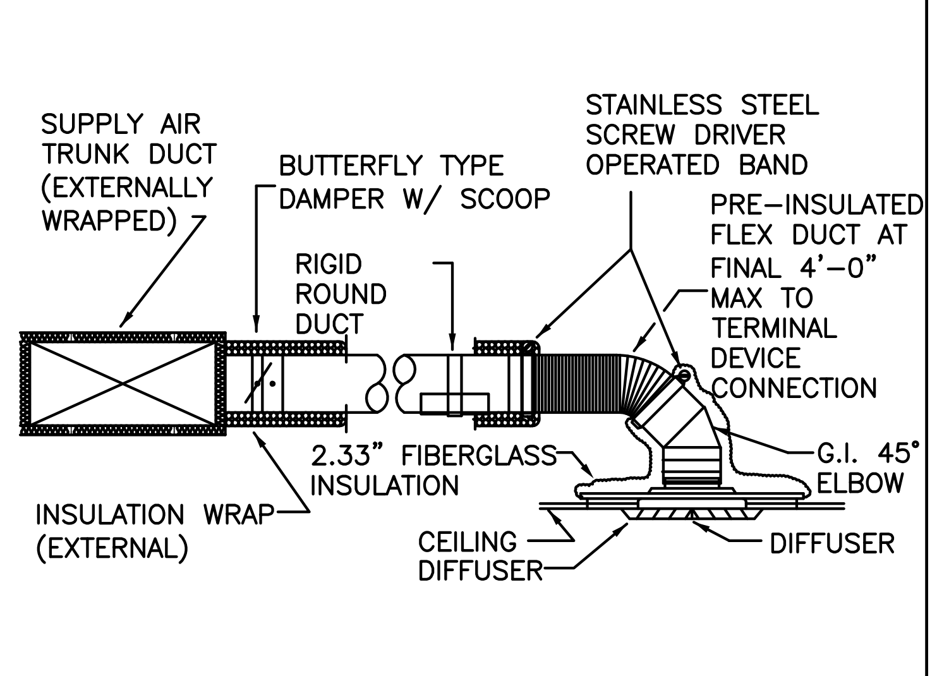

WALL OR GLASS
INSTALL DUCT CONNECTION PER DETAILS
INSULATE TOP OF DIFFUSER WITH 2.33" THICK FIBERGLASS INSULATION WITH FOIL BACKING AND SEAL WITH TAPE AROUND ALL FOUR SIDES OF DIFFUSER.
1/4 QUADRANT BLANK OFF
NO AIRFLOW FROM DIFFUSER IN THIS DIRECTION
CEILING ASSEMBLY
30" OR LESS
BLANK-OFF PLATE MOUNTED BEHIND DIFFUSER LOUVER FACE.
30" OR LESS
WALL
PLAN VIEW
ABOVE CEILING ISOMETRIC VIEW

NOTE: WHERE FINAL DIFFUSER LOCATION IS WITHIN 30" OR LESS OF A WALL SURFACE, SURFACE MOUNTED LIGHT FIXTURE, OR OTHER OBSTRUCTION IN AIR PATH, INSTALL QUADRANT BLANK-OFF TO LIMIT THROW TO DIRECTIONS WITH NO OBSTRUCTIONS.

1 DIFFUSER THROW CONTROL DETAIL

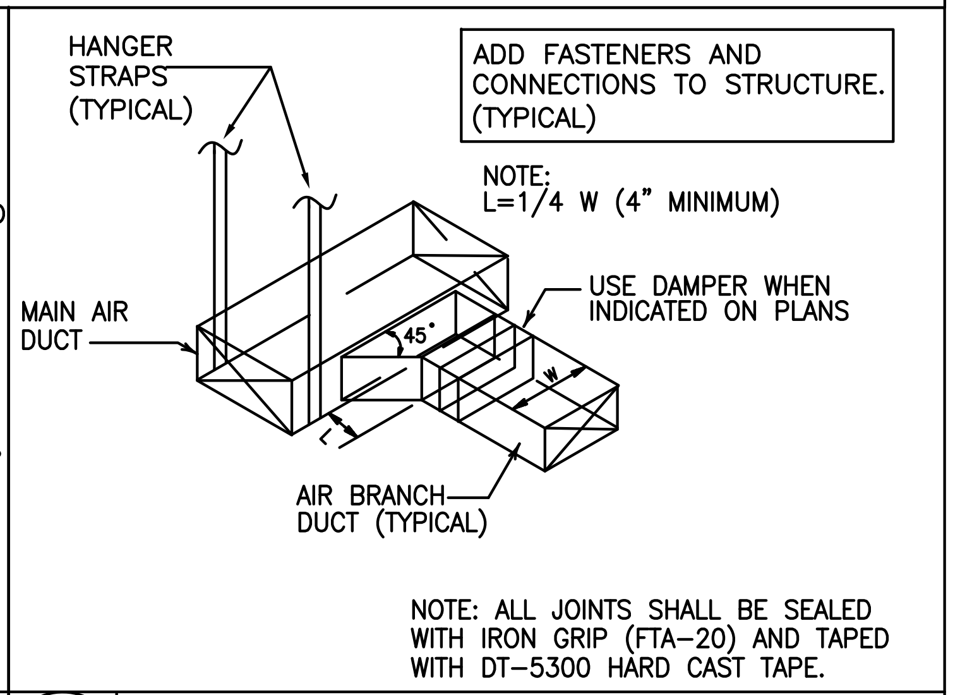
NO SCALE

(TYPICAL ALL OBSTRUCTION AREAS)


SUPPLY AIR TRUNK DUCT (EXTERNALLY WRAPPED)
BUTTERFLY TYPE DAMPER W/ SCOOP
RIGID ROUND DUCT
STAINLESS STEEL SCREW DRIVER OPERATED BAND
PRE-INSULATED FLEX DUCT AT FINAL 4'-0" MAX TO TERMINAL DEVICE CONNECTION
2.33" FIBERGLASS INSULATION
G.I. 45° ELBOW
INSULATION WRAP (EXTERNAL)
CEILING DIFFUSER
DIFFUSER

2 DIFFUSER / DUCT INSTALLATION DETAIL

NO SCALE



ADD FASTENERS AND CONNECTIONS TO STRUCTURE. (TYPICAL)

NOTE:
L=1/4 W (4" MINIMUM)

NOTE: ALL JOINTS SHALL BE SEALED WITH IRON GRIP (FTA-20) AND TAPED WITH DT-5300 HARD CAST TAPE.

3 AIR DUCT CONNECTION DETAIL

NO SCALE

ASSOCIATED DESIGN GROUP, INC.
A PROFESSIONAL ENGINEERING COMPANY

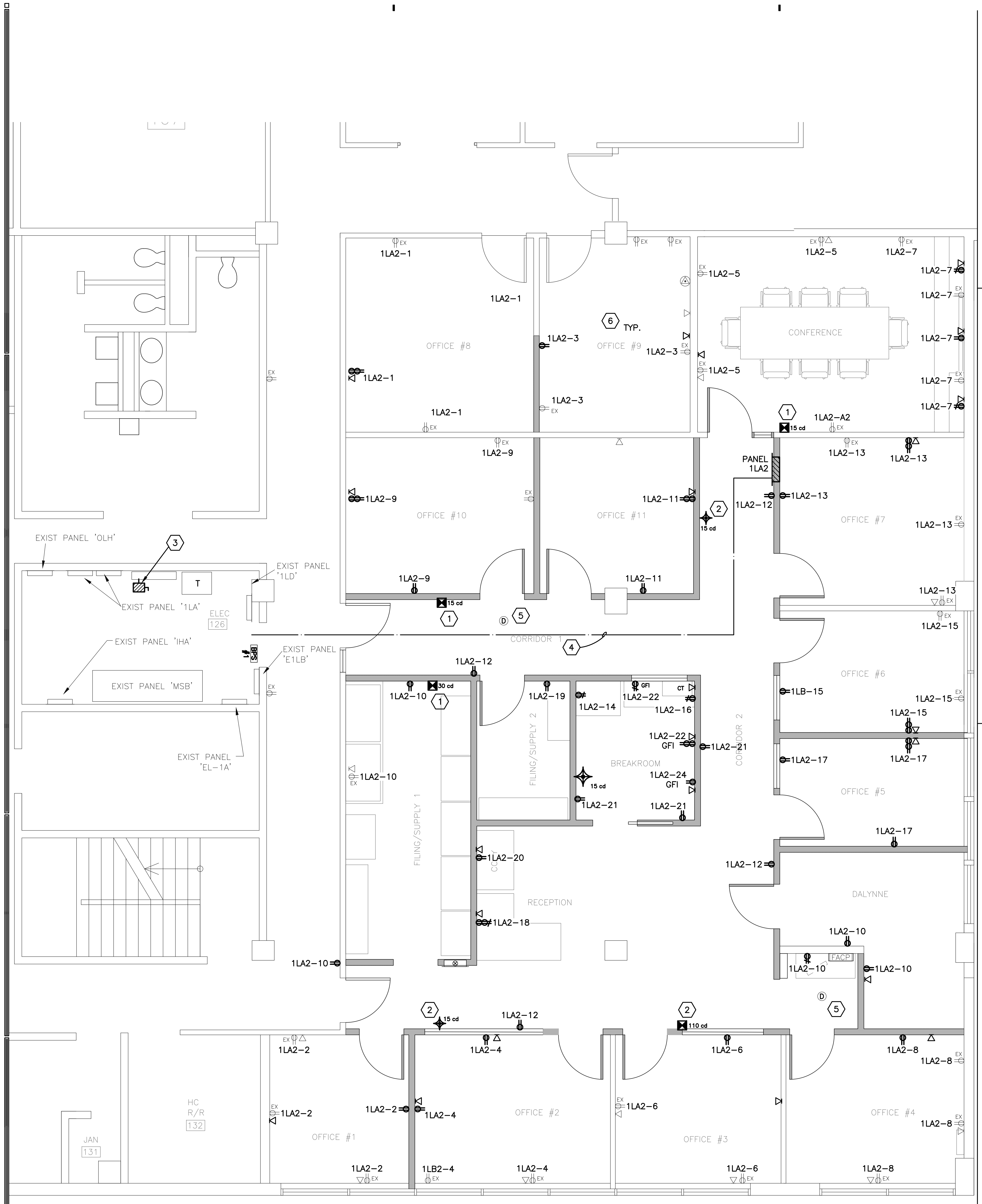
CORPORATE OFFICE
3909 West Congress Street
Suite 201
Lafayette, Louisiana 70506
VOICE (337) 234-5710
FAX (337) 237-1467
EMAIL: adginc@adginc.org

ADG Project No. 13124-P

UNITED STATES DEPARTMENT OF COMMERCE
NATIONAL OCEANIC & ATMOSPHERIC ADMINISTRATION
ATLANTIC OCEANOGRAPHIC AND METEOROLOGICAL LABORATORY
1ST FLOOR ADMINISTRATION RENOVATION
MIAMI, FLORIDA

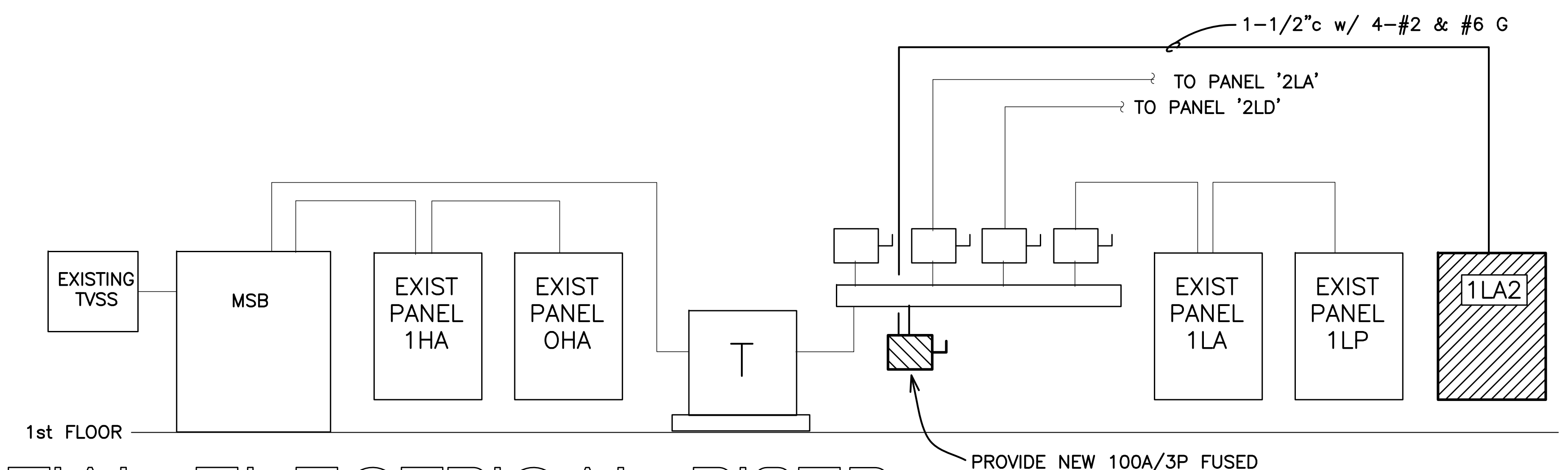
50% CD
DRAFT
SUBMITTAL

DATE ISSUED 09/??/15
REVISED
DRAWN BY JTP
CHECKED BY PB/LWB
PROJECT NO. 13124-P
SHEET NO. M1.0
OF



1ST FLOOR POWER & SPECIAL SYSTEMS PLAN

SCALE 1/4"=1'-0"



PARTIAL ELECTRICAL RISER

NO SCALE

GENERAL ELECTRICAL NOTES:

A. PROVIDE ALL LABOR AND MATERIALS FOR A COMPLETE FIRE ALARM SYSTEM.

B. RECONNECT EXISTING RECEPTACLES TO NEW CIRCUIT AS INDICATED. REPLACE EXISTING DEVICES WITH NEW RECEPTACLES.

C. WORK SHALL COMPLY WITH ALL LOCAL AND NATIONAL BUILDING CODES.

D. PROVIDE ADDITIONAL SUPPORTS AS REQUIRED FOR EXISTING CONDUITS, BOXES, ETC.

E. RECEPTACLES & DATA DEVICES REMOVED DURING DEMOLITION & SHOWN TO REMAIN TO BE RE-INSTQLLED & RECONNECTED AS REQUIRED.

ELECTRICAL KEYNOTES:

1. PROVIDE NEW FIRE ALARM DEVICE. CONNECT TO EXISTING FIRE ALARM SYSTEM.
2. INSTALL EXISTING DEVICE AND RE-CONNECT TO FIRE ALARM SYSTEM
3. NEW 100A/3P FUSED DISCONNECT SWITCH ATTACHED TO EXISTING WIREWAY. MAKE CONNECTION AS REQUIRED TO EXTEND SERVICE FROM WIREWAY TO DISCONNECT.
4. VERIFY EXISTING CONDITIONS TO FACILITATE ROUTING OF FEEDER FROM ELECTRICAL ROOM TO NEW PANEL.
5. REUSE EXISTING SMOKE DETECTORS. INSTALL AS SHOWN IN-LIEU OF PROVIDING NEW DEVICE. COORDINATE EXACT LOCATION WITH GOVERNMENT REPRESENTATIVE.
6. ADJACENT EXISTING DEVICE MOUNTING HEIGHT TO MATCH.

PANEL DESCRIPTION: **1LA2**

MOUNTING: RECESSED
A.I.C.: 10,000
MAIN: 100A MC.B.
VOLT: 120/208
PHASE: 3
WIRE: 4 + GND

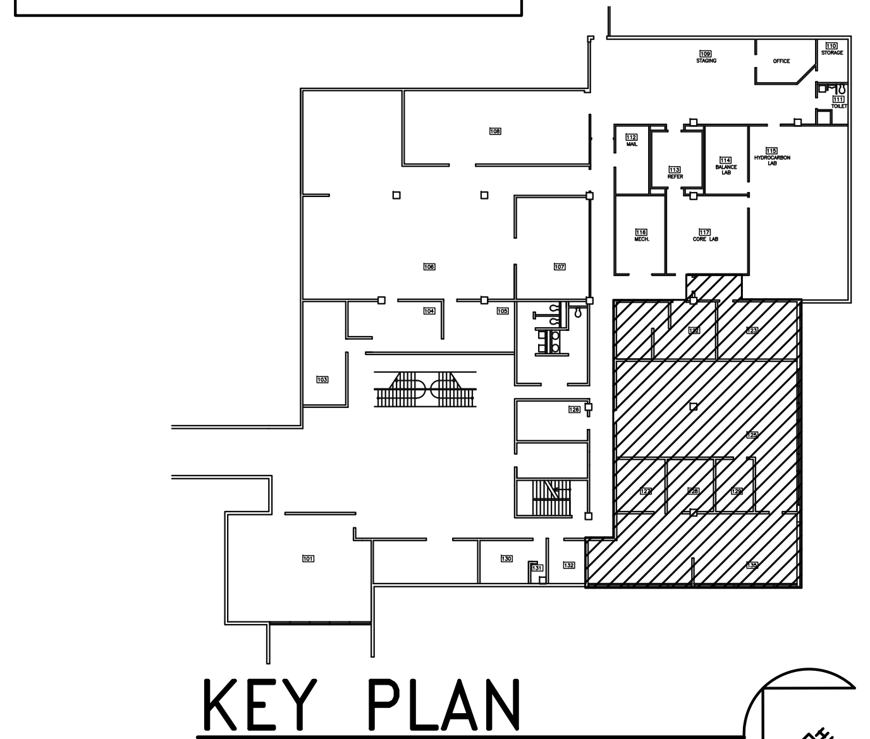
SERVING	CKT #	BKR AMP.	WIRE	GND	COND.	PHASE A B C	COND.	GND	WIRE	BKR AMP.	CKT #	SERVING
RECEPT. (OFFICE)	1	20	2-12	12	3/4"	A	3/4"	12	2-12	20	2	RECEPT. (OFFICE)
RECEPT. (OFFICE)	3	20	2-12	12	3/4"	B	3/4"	12	2-12	20	4	RECEPT. (OFFICE)
RECEPT. (CONF.)	5	20	2-12	12	3/4"	C	3/4"	12	2-12	20	6	RECEPT. (OFFICE)
RECEPT. (CONF.)	7	20	2-12	12	3/4"	A	3/4"	12	2-12	20	8	RECEPT. (OFFICE)
RECEPT. (OFFICE)	9	20	2-12	12	3/4"	B	3/4"	12	2-12	20	10	RECEPT. (FILING)
RECEPT. (OFFICE)	11	20	2-12	12	3/4"	C	3/4"	12	2-12	20	12	RECEPT. (COMMONS)
RECEPT. (OFFICE)	13	20	2-12	12	3/4"	A	3/4"	12	2-12	20	14	RECEPT. (FRIDGE)
RECEPT. (OFFICE)	15	20	2-12	12	3/4"	B	3/4"	12	2-12	20	16	RECEPT. (MICROWAVE)
RECEPT. (OFFICE)	17	20	2-12	12	3/4"	C	3/4"	12	2-12	20	18	RECEPT. (RECEPTION)
RECEPT. (OFFICE)	19	20	2-12	12	3/4"	A	3/4"	12	2-12	20	20	RECEPT. (COPY)
RECEPT.	21	20	2-12	12	3/4"	B	3/4"	12	2-12	20	22	VENDING
RECEPT.	23	20	2-12	12	3/4"	C	3/4"	12	2-12	20	24	VENDING
SPARE C.B.	25	20	–	–	–	A	–	–	–	20	26	SPARE C.B.
SPARE C.B.	27	20	–	–	–	B	–	–	–	20	28	SPARE C.B.
SPARE C.B.	29	20	–	–	–	C	–	–	–	20	30	SPARE C.B.
SPARE C.B.	31	20	–	–	–	A	–	–	–	20	32	SPARE C.B.
SPARE C.B.	33	20	–	–	–	B	–	–	–	20	34	SPARE C.B.
SPARE C.B.	35	20	–	–	–	C	–	–	–	20	36	SPARE C.B.
SPARE C.B.	37	20	–	–	–	A	–	–	–	20	38	SPARE C.B.
SPARE C.B.	39	20	–	–	–	B	–	–	–	20	40	SPARE C.B.
SPARE C.B.	41	20	–	–	–	C	–	–	–	20	42	SPARE C.B.

DEMAND LOAD: øA = 8.2K øB = 7.6K øC = 7.6K

TOTAL LOAD 23.4 KVA
65.0 AMPS

NOTE:
(1) PROVIDE GROUND BUS.
(2) PROVIDE GROUND CONDUCTOR IN EACH.
(3) PROVIDE HINGED COVER AND HINGED TRIM.
(4) NEUTRAL CONDUCTORS SHALL NOT BE SHARED.

NOTE:
REFER TO ARCHITECTURAL PLANS FOR ALL DIMENSIONS.



KEY PLAN
NOT TO SCALE

Associated Design Group, Inc.
A PROFESSIONAL ENGINEERING COMPANY

CORPORATE OFFICE
3909 West Congress Street
Suite 201
Lafayette, Louisiana 70506
VOICE (337) 234-5710
FAX (337) 237-1467
EMAIL: adginc@adginc.org

ADG Project No. 13124-P

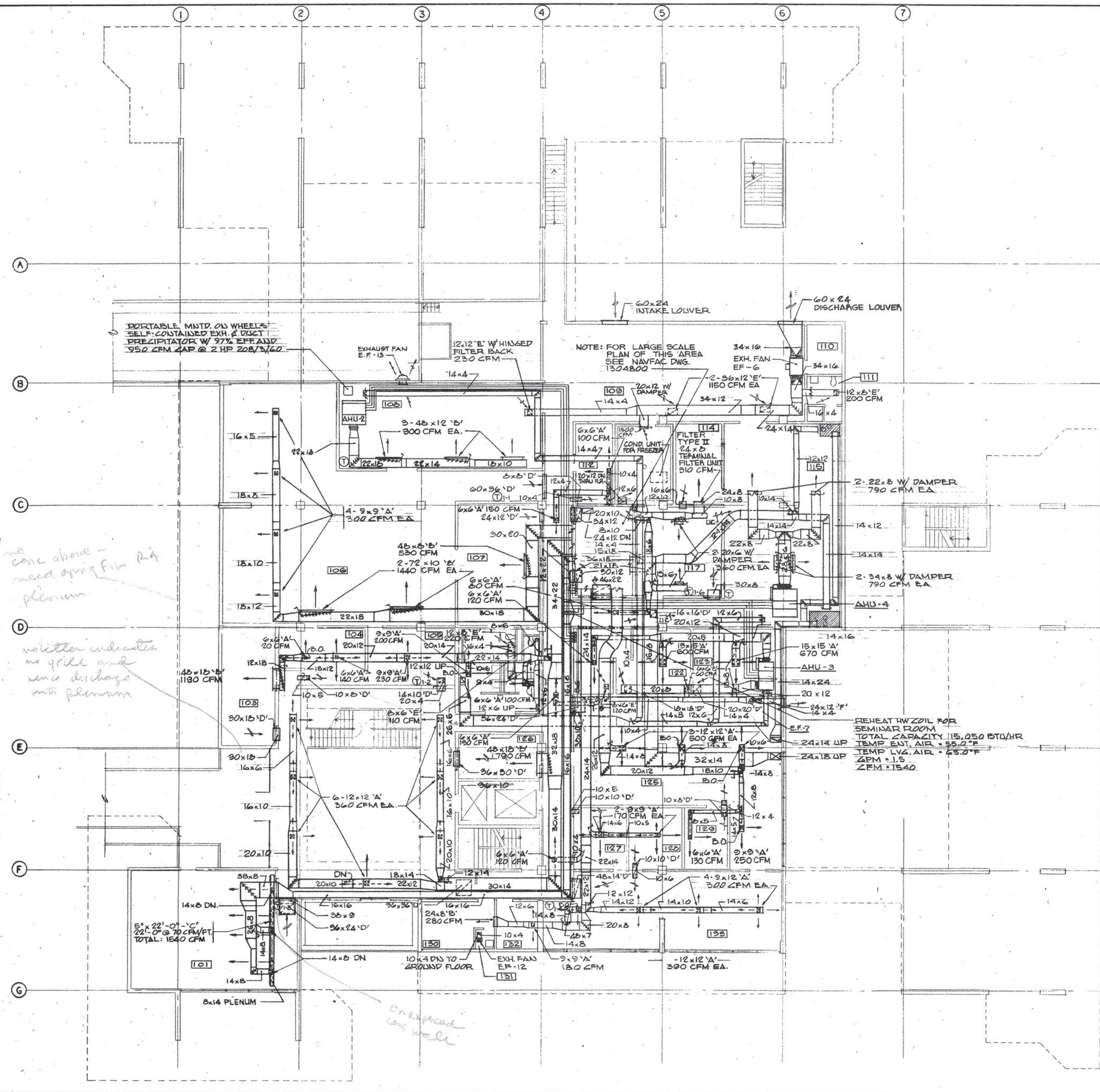
UNITED STATES DEPARTMENT OF COMMERCE
NATIONAL OCEANIC & ATMOSPHERIC ADMINISTRATION
ATLANTIC OCEANOGRAPHIC AND METEOROLOGICAL LABORATORY
1ST FLOOR ADMINISTRATION RENOVATION
MIAMI, FLORIDA

VOICE/DATA
REVISIONS

DATE ISSUED 04/12/16
REVISED Nov. 22, 2016
DRAWN BY KS
CHECKED BY LM
PROJECT NO. 13124-P
SHEET NO. E1.1
OF

L:\2013\13124-P AOML MIAMI 1st Fl Adm Renov\Drawings\13124-P E1.1.dwg, E1.1, 12/5/2016 11:23:26 AM, dannyh, Adobe PDF

LET	DESCRIPTION	PREP'D BY	DATE	APPROVED
REVISIONS				

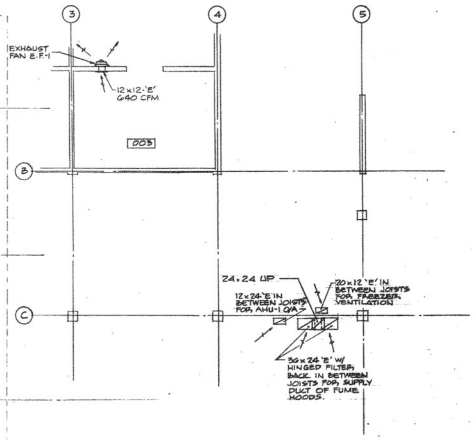


no cone above – need opng for RA plenum

no letter indicates no grill and hence discharge into plenum

Enclosed cone wall

NOTE 1.
SEE NAVFAC DRWG 1304805
FOR SYMBOL LEGEND


3
4
5
B
C
EXHAUST FAN E.F.-1
12x12 'E' 640 CFM
003
24x24 UP
20x12 'E' IN BETWEEN JOISTS FOR FREEZER VENTILATION
12x24 'E' IN BETWEEN JOISTS FOR AHU-1 O/A
36x24 'E' W/ HINGED FILTER BACK IN BETWEEN JOISTS FOR SUPPLY DUCT OF FUME HOODS.

PARTIAL GROUND FLOOR
SCALE: 1/8" = 1'0"

GRAPHIC SCALE
1/8" = 1'-0" 5' 0 5' 10' 15' 20' 25'

NORTH

FIRST FLOOR PLAN
SCALE 1/8" = 1'-0"

PANCOAST/FERENDINO/GRAFTON ARCHITECTS/ENGINEERS 2575 S. BAYSHORE DRIVE MIAMI-FLORIDA 33133	DEPARTMENT OF THE NAVY NAVAL FACILITIES ENGINEERING COMMAND SOUTHEAST DIVISION CHARLESTON, S. C.			
DSGN VALLE DR BERM CHK DUMOIS	VIRGINIA KEY MIAMI-FLORIDA ENVIRONMENTAL SCIENCE SERVICE ADMINISTRATION ATLANTIC OCEANOGRAPHIC LABORATORIES FIRST FLOOR PLAN - AIR COND.			
SUPV VALLE CH ENGR MARTINEZ				
SUBMITTED BY P. SPILLIS PARTNER DATE SEP. 26/69 FIRM MEMBER (TITLE)				
SOEAST DIV-DESIGN DIVISION PDE Wilson DIR C.D. Black				
APPROVED DATE OFFICER IN CHARGE	SIZE F	CODE IDENT NO. 80091	NAVFAC DRAWING NO. 1304794	
APPROVED DATE 3/25/70 FOR COMMANDER, NAVFAC			CONSTR CONTR NO. N62467-68-C-0081	
	SCALE AS SHOWN		SPEC 91191/67	SHEET 61 OF 99

AC-1

REDUCED DRAWING USE GRAPHIC SCALE